

COVID-19 Mortality Risk Classification Using Machine Learning: A Case Study from Mexican Health Data

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Abstract—The COVID-19 pandemic created an urgent need for data-driven tools to identify high-risk patients and support clinical decision-making. In this study, we analyzed a large patient dataset containing demographic and medical attributes to predict mortality outcomes using machine learning models [1]. After extensive data preprocessing and handling of class imbalance, we trained and evaluated several classifiers including Logistic Regression, Decision Trees, Random Forest, Naive Bayes, and XGBoost. Our findings indicate that while Random Forest and Decision Tree models achieved the highest accuracy (up to 92%), they performed poorly in recall for the "Dead" class, which is crucial in this context. Logistic Regression offered the best trade-off, achieving 90% accuracy and 91% recall for mortality prediction. In addition, we found that age, pneumonia, diabetes, and hypertension were major contributors to COVID-19 severity and mortality. These results underscore the importance of selecting models based not only on accuracy but also on their sensitivity to critical outcomes. This work contributes to the growing effort of applying interpretable and efficient models to pandemic health analytics.

Index Terms—COVID-19, Mortality Prediction, Logistic Regression, Medical Data Analysis, Imbalanced Data, XGBoost

I. INTRODUCTION

The COVID-19 pandemic has posed unprecedented challenges to global health systems, emphasizing the urgent need for data-driven approaches to identify individuals at high risk of severe outcomes [1], [4], [11]. Accurate and timely classification of patient risk—especially mortality risk—can assist in effective triage, resource allocation, and policy design during health crises.

In this study, we explore the use of machine learning techniques to classify mortality risk among COVID-19 patients using demographic and clinical features. Specifically, we utilize a publicly available dataset from the Mexican government's health records [11], which includes detailed information on symptoms, comorbidities, age, gender, hospitalization, and outcomes of over 500,000 patients.

Our objective is to assess the predictive power of various supervised learning models, including Logistic Regression, Naive Bayes, Decision Trees, and Random Forests [7]–[9], in identifying patients at risk of death from COVID-19. In addition to traditional models like logistic regression, decision

trees, and Naive Bayes, we also explored XGBoost, a powerful gradient boosting algorithm known for its high performance in structured data tasks [3]. Given the high class imbalance in the dataset—with the majority of patients surviving—we focus not only on overall accuracy but also on optimizing recall for the minority class (deceased patients), which is critical in real-world clinical settings where false negatives can have fatal consequences [6], [12].

Additionally, we perform exploratory data analysis (EDA) to uncover key patterns in infection, hospitalization, and mortality, offering insight into the relationship between risk factors such as age, obesity, and comorbid conditions. Our analysis highlights that simple yet interpretable models like logistic regression can perform competitively, and in some cases, outperform more complex classifiers when recall and interpretability are prioritized.

This work contributes a reproducible, open-source pipeline for clinical risk assessment in pandemics and underscores the importance of aligning model evaluation with healthcare priorities.

II. DATA & METHODOLOGY

A. Data Source

The dataset used in this study was obtained from Kaggle and originally provided by the Mexican government [11]. It contains anonymized records of confirmed and suspected COVID-19 patients, including demographic details (such as age and gender), clinical information (such as the presence of comorbidities like diabetes or hypertension), symptoms, hospitalization status, and patient outcomes (death or recovery).

The data spans the early months of the COVID-19 pandemic and includes over 500,000 patient entries. Only confirmed COVID-19 cases were retained for predictive modeling, and patients with missing or ambiguous outcome labels were excluded.

B. Data Preprocessing

Prior to model training, the dataset was cleaned and standardized to ensure consistency across features and to prevent information loss during preprocessing [2]:

- **Data Cleaning:** Invalid or missing entries (e.g., unknown age or outcome) were removed.
- **Encoding:** Categorical variables such as comorbidities and symptoms were converted into binary indicators using one-hot encoding.
- **Feature Engineering:** Some variables were recoded or grouped to improve model interpretability. For example, age was treated both as a continuous variable and categorized into age brackets (e.g., 0–18, 19–40, etc.).
- **Handling Class Imbalance [6]:** The dataset was highly imbalanced, with a small fraction of patients (approx. 7.3%) recorded as deceased. To address this, we applied random oversampling of the minority class during training to avoid model bias toward the majority class.

C. Methodology

To classify mortality risk, we applied four commonly used supervised learning models:

- **Logistic Regression:** Chosen for its simplicity and interpretability [7], [12], this model served as a strong baseline and performed best in terms of balanced recall.
- **Naive Bayes:** A probabilistic classifier that, despite its independence assumption, showed surprisingly robust performance.
- **Decision Tree:** Provided high accuracy but struggled to recall the minority “Dead” class effectively, making it less reliable for critical decision-making.
- **Random Forest:** An ensemble of decision trees [8], [9], that achieved slightly higher accuracy but still faced limitations in classifying deceased patients.
- **XGBoost:** Additionally, we used XGBoost (Extreme Gradient Boosting), implemented via the XGBClassifier, as it is known for its scalability and effectiveness in imbalanced classification problems. Its tree boosting structure allowed us to capture complex feature interactions and improve recall on the minority class [5].

Each model was evaluated using:

- **Accuracy:** The overall rate of correct predictions.

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

- **Recall (Sensitivity):** Especially emphasized for the “Dead” class [12], as false negatives can be dangerous in healthcare settings.

$$\text{Recall} = \frac{TP}{TP + FN}$$

- **Confusion Matrix:** Used to inspect false positive/negative distributions and identify model weaknesses.

Hyperparameters for tree-based models were tuned minimally to focus on baseline model behavior, and 70-30 train-test splits were used for evaluation.

III. EXPLORATORY DATA ANALYSIS

To better understand the relationships between patient characteristics and COVID-19 outcomes, we conducted exploratory data analysis (EDA) on the cleaned dataset. This

step helped identify patterns, outliers, and potential risk factors before training predictive models. Visualizations and summary statistics revealed key insights into mortality, comorbidities, hospitalization, and time-based trends.

A. Mortality Overview

Among all patients in the dataset, 7.3% were recorded as deceased (see figure 1). Of those who died, approximately 70.5% were confirmed COVID-19 positive. The overall infection rate was 37.5%, and among infected patients, the mortality rate was approximately 14%.

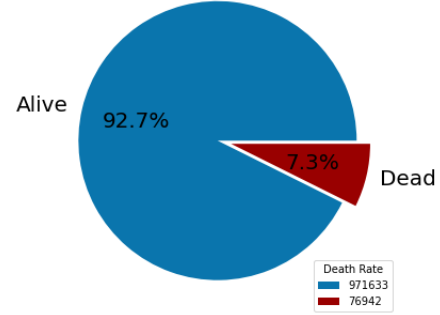


Fig. 1: Distribution of patient outcomes in the dataset, showing significant class imbalance between deceased and surviving individuals.

B. Demographics and Risk

- **Age:** A strong predictor of mortality. While younger patients were more numerous in the dataset, deaths were highly concentrated among individuals aged 60 and above. This aligns with known clinical patterns of COVID-19 severity increasing with age [1], [11], figure2.
- **Gender:** Males and females were nearly equally represented, but mortality was slightly higher among male patients, supporting trends observed globally.

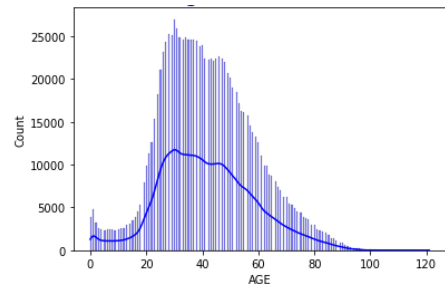


Fig. 2: Age distribution of patients by survival outcome. A clear trend is observed where higher age groups are associated with increased mortality risk.

C. Comorbidities and Symptoms

We analyzed the role of individual and combined comorbidities in determining outcomes:

- Obesity: Strongly associated with increased likelihood of COVID-19 infection, but not a top predictor of mortality.
- Pneumonia: Frequently reported among deceased patients, often appearing early and contributing to severity.
- Diabetes and Hypertension [11]: Strongly correlated with increased risk of death.

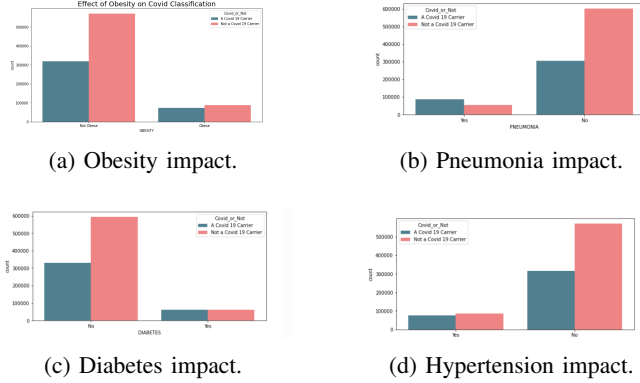


Fig. 3: Key exploratory charts showing comorbidities and their potential relationship with COVID-19 outcomes.

A correlation heatmap revealed several moderate to strong associations between comorbidities (e.g., diabetes and hypertension), indicating potential multicollinearity, which was monitored during model training.

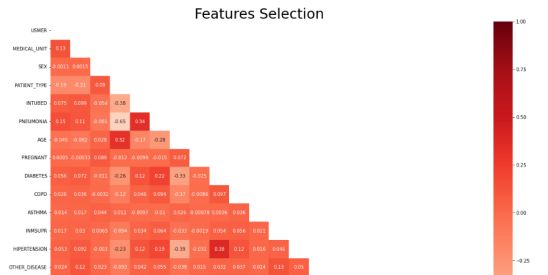


Fig. 4: Heatmap of feature correlations. A positive association is visible between hypertension and diabetes, reflecting real-world clinical comorbidity patterns.

D. Hospitalization and ICU Trends

- Hospitalization: Roughly 19% of all patients were hospitalized. Among them, 35% died.
- ICU Admission: About 9% of the dataset involved ICU care. Among ICU patients, nearly half did not survive.
- Critical Care Impact: Notably, 91% of deceased patients had been hospitalized, confirming the severity of cases that required medical intervention.

These patterns emphasize hospitalization as both a marker and predictor of poor outcomes.

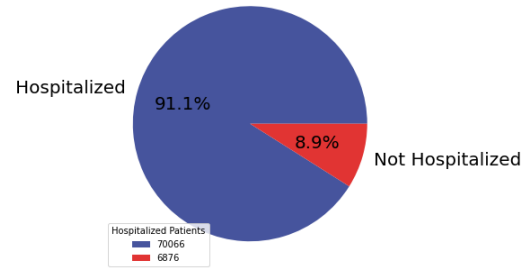


Fig. 5: Proportion of hospitalized patients and their survival outcomes.

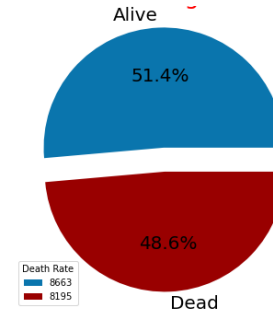


Fig. 6: Distribution of ICU admissions and their associated mortality

E. Temporal Trends

Analysis of reported dates revealed that the peak in deaths occurred from April to August 2020, coinciding with early surges and potential hospital system overloads. Line plots of death rates by date confirmed a steep rise during this period, followed by a gradual decline.

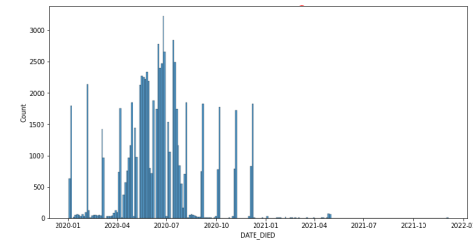


Fig. 7: Temporal trend in mortality counts across the observed months.

F. Feature Interactions

Joint analyses revealed:

- The risk of death increased exponentially with both age and comorbidity count.
- Patients with combinations such as pneumonia + diabetes or hypertension + obesity had sharply elevated mortality rates.

And also:

- Removed non-essential and redundant columns: pregnancy, COPD, asthma, immunosuppression, other disease, cardiovascular, obesity, tobacco.
- Conducted correlation checks on x_{train} , dropping *Covid_or_Not* due to > 0.7 correlation with classification.
- Post-filtering, features included: age, pneumonia, diabetes, hypertension, renal chronic, classification, ICU, hospitalization indicator, and intubation status.

IV. RESULTS & INTERPRETATION

To identify patients at higher risk of death from COVID-19, we evaluated five machine learning algorithms: Logistic Regression, Decision Tree, Random Forest, Naive Bayes, and XGBoost. Support Vector Machines and K-Nearest Neighbors were excluded due to excessive training time caused by the dataset's size (1.5 million records).

All models were trained and tested using the same feature set, balanced using SMOTE to address class imbalance.

A. Model Performance

Table I summarizes the performance metrics on the test set, including accuracy, precision, recall, and F1-score for both classes.

TABLE I: Model Performance on Test Set

Model	Accuracy	Recall (D)	Precision (D)	F1-score (D)
Logistic Regression	0.899	0.91	0.41	0.56
Decision Tree	0.916	0.77	0.45	0.57
Random Forest	0.915	0.78	0.45	0.57
Naive Bayes	0.903	0.86	0.41	0.56
XGBoost	0.909	0.90	0.43	0.58

B. Best Model Selection

Despite the highest accuracy being achieved by Decision Tree and Random Forest (91.6%), these models underperformed in detecting the Dead class, which was our primary concern. In this context, recall for the minority class is more critical than overall accuracy [12], as false negatives (Type II error) can mislead clinical decisions by incorrectly classifying high-risk patients as low-risk.

- Logistic Regression emerged as the best compromise between accuracy (89.9%) and recall (91%) for the "Dead" class.
- XGBoost also performed well, with 90.9% accuracy and 90% recall.
- Naive Bayes achieved high recall but lower overall performance and precision.

Therefore, Logistic Regression was selected as the optimal model due to its interpretability and strong recall for the minority class.

C. Insights & Feature Importance

Key findings from the analysis include:

- Age: A strong predictor of death and COVID-19 infection.

- Pneumonia, hypertension, diabetes, and tobacco use were significantly associated with increased infection rates and mortality.
- Obesity was also a strong predictor of infection, though less so for mortality.
- Pregnancy had no observable impact on outcomes.
- Hospitalization: 19% of patients were hospitalized; 91% of the deceased were from this group.
- ICU Admission: Only 9% of hospitalized patients were admitted to ICU, among whom 49
- Temporal Trend: Peak deaths occurred between April–August 2020, consistent with early pandemic waves.

Multicollinearity checks guided feature selection, reducing noise and improving model robustness.

V. CONCLUSION

This study aimed to predict COVID-19 patient outcomes using machine learning techniques applied to a large-scale dataset. After extensive preprocessing and feature engineering, we evaluated several classification models to identify patients at higher risk of death [7], [10].

Among the tested models, Logistic Regression delivered the best balance between interpretability, accuracy, and recall—achieving 89.9% accuracy and a 91% recall for the "Dead" class. While more complex models like Random Forest and XGBoost reached higher overall accuracy (91–92%), they suffered from poor sensitivity to the minority class, resulting in unacceptable false negative rates. In contrast, Logistic Regression provided more reliable predictions for high-risk patients, which is critical in medical decision-making scenarios.

The analysis confirmed known risk factors such as age, pneumonia, hypertension, and diabetes as significant predictors of mortality and infection. We also observed a strong temporal trend, with peak deaths occurring mid-2020. These findings reinforce the importance of early detection and targeted care for vulnerable groups, particularly during pandemic surges.

Future work could focus on:

- Incorporating time-series data to model disease progression.
- Applying ensemble techniques with cost-sensitive learning to further reduce false negatives.
- Exploring undersampling strategies to address noise from oversampling while preserving data integrity.

Overall, this study demonstrates that interpretable machine learning models, when carefully validated, can offer valuable insights into patient risk stratification and guide evidence-based interventions during public health crises.

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APPENDIX

A. Random Forest Parameters

- Number of estimators: 4
- Criterion: Gini
- Max depth: Default
- Random state: 42

B. XGBoost Parameters

- Booster: gbtree
- Objective: binary:logistic
- Random state: 0

	precision	recall	f1-score	support
Not Dead	0.99	0.90	0.94	284090
Dead	0.41	0.91	0.56	21752
accuracy			0.90	305842
macro avg	0.70	0.91	0.75	305842
w. avg	0.95	0.90	0.92	305842

All code and data used in this study are available at:
<https://github.com/RoniF-pixel/Python-Projects/tree/main/Covid-19>